Lecture 38

Double integrals and the Gaussian

Uses.

Summary 5: Function limits obey arithmetic, order and squeeze laws.
Summary 8: Scalar differentiation and the trigonometric derivative rules hold.
Summary 14: Polar coordinates give $x = r\cos\varphi$, $y = r\sin\varphi$ with $r \geq 0$ and an angle modulo 2π.
Summary 21: The absolute determinant is the area factor of a linear map in the plane.
Summary 26: Proper continuous integrals obey finite linearity and order; primitives, substitution and parts evaluate them.
Summary 27: Improper integrals require separate endpoint limits. Both Gaussian half-lines converge, and $\Gamma(1/2) = 2\int_0^\infty e^{-x^2}\,\mathrm{d}x$. The error function is a finite Gaussian integral.
Summary 11: The scalar finite Taylor formula bounds the quadratic approximation by its third derivative at an intermediate point.
Summary 36: The Jacobian matrix represents the derivative, and Jacobians compose by matrix multiplication.

Objectives. *A double integral sums a quantity spread over a planar region. We state the compact-domain integration rules used in this course, then use region geometry and Jacobians to evaluate those integrals. The Gaussian value follows by comparing compact discs and squares before taking ordinary improper limits. The same integral notation gives mass and the centroid of a plate with varying density.*

By the end of this lecture you will be able to:

- State standing fact M with its continuity, compact-region and change-of-variables hypotheses and its restriction to proper integrals.
- Rewrite an iterated integral by describing the same region in the opposite order.
- Compute the absolute Jacobian factor and apply admissible planar changes of variables, including polar coordinates.
- Prove the Gaussian value through compact-domain bounds and compute planar mass and first moments.

38.1 Small-area sums

On a rectangle $D = [a, b] \times [c, d]$, partition both coordinate intervals and choose a point in each small rectangular cell. The associated *double Riemann sum* is

$$\sum_{i,j} f(\xi_{ij}, \eta_{ij})\, \Delta x_i \Delta y_j.$$

The factor $\Delta x_i \Delta y_j$ is the area of that cell. A double integral is the common limit of these sums as the largest cell diameter tends to zero, independently of the sample points. For a region partitioned into small-area pieces D_k, the corresponding sum is $\sum_k f(\xi_k)\,\mathrm{area}(D_k)$ with $\xi_k \in D_k$. We write the limit as $\iint_D f \,\mathrm{d}A$. Existence and the integration rules below are supplied by standing fact M for the continuous integrands and regions used here.

A region of *vertical graph type* has the form

$$D = \{(x, y) : a \le x \le b,\ g_1(x) \le y \le g_2(x)\},$$

with continuous $g_1 \le g_2$. Horizontal graph type reverses the coordinate roles. We also allow finite unions of such pieces. The boundaries in the standing fact are piecewise C^1 curves, meaning finite unions of C^1 parametrised curve pieces. A circle qualifies even where its upper and lower graph derivatives become unbounded. The regions are closed and bounded; these are the compact regions meant in the integration statements here.

Standing fact M. We use the following proper multiple-integral rules in two and three dimensions. The parameter-differentiation clause also covers proper integrals over a compact interval. The integrand is continuous on a compact region between graphs, or a finite union of such regions, with piecewise C^1 boundary. The corresponding compact graph regions and boundary conditions for three-dimensional solids are stated in Lecture 39 before their use.

- Such multiple integrals exist. They equal the iterated integrals in any order for which the region is described by admissible graph pieces. This is the compact-region *Fubini rule*.
- Let a C^1 map Φ on a neighbourhood of a compact parameter region E map E onto D. If it is injective on the interior and has $\det J_\Phi \neq 0$ there, change of variables multiplies the integrand by $|\det J_\Phi|$. Failures of injectivity or nonsingularity on the boundary, such as a polar origin or seam, are allowed.
- For a fixed compact integration region, differentiating a proper integral with respect to a parameter is allowed when the integrand is C^1 jointly on a neighbourhood of the compact parameter–integration product. The derivative is the integral of the corresponding parameter derivative, at interior parameter values.

These are standing facts, rather than theorems proved in this course. Their sources are Zorich, Volume II, Chapter 11 and §17.1, and Stewart, §15.9. Results whose only unproved input is M count as proved under the course's standing-fact policy. M makes no assertion about an improper multiple integral. Every unbounded-region limit below is justified separately.

Finite sums are linear in the integrand and preserve order. Passing to their defining limits gives the same properties for the integral. Partitioning a region into finitely many pieces with disjoint interiors gives additivity. In particular, for a nonnegative continuous integrand, $D_1 \subset D_2$ implies $\iint_{D_1} f \le \iint_{D_2} f$ when the closure of the difference is a finite union of admissible pieces. For $f = 1$ the integral is the region's area. For a bounded integrand this also gives $|\iint_D f| \le \sup_D |f|\,\mathrm{area}(D)$.

38.2 Fubini and the order of integration

For a vertical graph region, M gives the explicit formula

$$\iint_D f \, dA = \int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\, dy \right) dx.$$

In the inner integral x is fixed. Its value becomes the function integrated with respect to x. Reversing the order requires new bounds describing the same points, sometimes in several graph pieces. For example, on $[0,1]\times[0,2]$,

$$\iint_D (x+2y)\, dA = \int_0^1 (2x+4)\, dx = 5.$$

The other order gives $\int_0^2 (1/2+2y)\, dy = 5$. Both calculations use M on the same compact rectangle and a continuous polynomial.

Let $D = \{(x,y) : 0 \le y \le x \le 1\}$. Figure 38.1 shows a vertical and a horizontal description of this triangle.

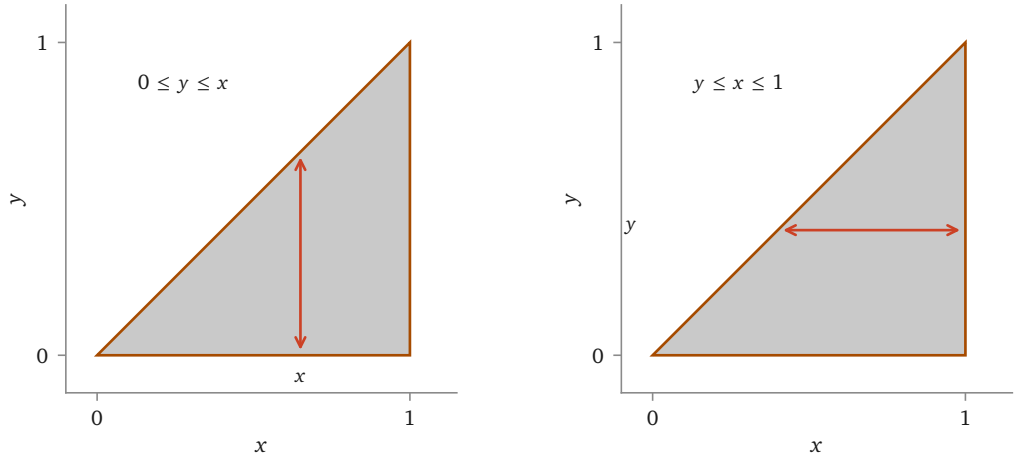


Figure 38.1: The same triangle described by vertical slices $0 \le y \le x$ and horizontal slices $y \le x \le 1$. The outer coordinate ranges from 0 to 1 in either description.

Example 38.1 (Choosing the order on a triangle)**.** Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$. Its region is exactly the triangle just described, and the integrand is continuous. M allows the reversed order, whose inner integrand is constant in y:

$$\int_0^1 \int_y^1 e^{x^2}\, dx\, dy = \int_0^1 \int_0^x e^{x^2}\, dy\, dx = \int_0^1 x e^{x^2}\, dx = \frac{e-1}{2}.$$

The last equality follows from the primitive $e^{x^2}/2$. The bounds changed because the slice direction changed; exchanging only the two differential symbols would describe a different integral.

38.3 Change of variables and the absolute determinant

For a planar transformation $(x,y) = \Phi(u,v)$ satisfying M, the rule is

$$\iint_D f(x,y)\, dx\, dy = \iint_E f(\Phi(u,v))\, |\det J_\Phi(u,v)|\, du\, dv. \tag{38.1}$$

The derivative gives the local linear map, whose area factor is the absolute determinant by Lecture 21. Standing fact M asserts that these local factors give the integral formula under the stated hypotheses. This is not a derivation of M itself. The absolute value keeps area positive even when the map reverses orientation.

Example 38.2 (An orientation-reversing coordinate change)**.** Evaluate $\iint_D (x^2 - y^2)^2 \, dA$ on $D = \{(x, y) : |x + y| \le 2,\ |x - y| \le 1\}$. Use $u = x + y$ and $v = x - y$, whose inverse map is

$$\Phi(u, v) = \left(\frac{u+v}{2}, \frac{u-v}{2}\right), \qquad J_\Phi = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}, \quad \det J_\Phi = -\frac{1}{2}.$$

The parameter region is $E = [-2, 2] \times [-1, 1]$. The linear map is a bijection onto D and is C^1 with nonzero determinant everywhere. Since $x^2 - y^2 = uv$, the change and Fubini rules give

$$\iint_D (x^2 - y^2)^2 \, dA = \frac{1}{2} \int_{-2}^{2} u^2 \, du \int_{-1}^{1} v^2 \, dv = \frac{1}{2} \frac{16}{3} \frac{2}{3} = \frac{16}{9}.$$

As a geometric check, the area of D is $(1/2)(4)(2) = 4$. Using the signed determinant in place of its absolute value would give a negative value for a nonnegative integrand.

38.4 Polar integration on compact regions

The polar map is $\Phi(r, \varphi) = (r \cos \varphi, r \sin \varphi)$. Its Jacobian is

$$J_\Phi = \begin{pmatrix} \cos \varphi & -r \sin \varphi \\ \sin \varphi & r \cos \varphi \end{pmatrix}, \qquad \det J_\Phi = r(\cos^2 \varphi + \sin^2 \varphi) = r.$$

On a single angular sweep of width at most 2π and positive interior radius, the map is injective in the parameter interior. The origin $r = 0$ and the two edges of a full turn are parameter-boundary exceptions expressly allowed by M. Thus the area element is $dA = r \, dr \, d\varphi$. For the disc $D_R = \{(x, y) : x^2 + y^2 \le R^2\}$, with $R > 0$,

$$\iint_{D_R} f \, dA = \int_0^{2\pi} \int_0^R f(r \cos \varphi, r \sin \varphi) r \, dr \, d\varphi.$$

For $f = 1$ this gives $2\pi R^2/2 = \pi R^2$, agreeing with disc area. For the continuous radial Gaussian it gives the compact-domain value

$$\iint_{D_R} e^{-(x^2+y^2)} \, dA = 2\pi \int_0^R r e^{-r^2} \, dr = \pi(1 - e^{-R^2}). \tag{38.2}$$

The final primitive is $-e^{-r^2}/2$. Only finite radii and proper integrals have been used.

38.5 The Gaussian value from compact bounds

Let $J_R = \int_{-R}^{R} e^{-x^2} \, dx$. On the compact square $Q_R = [-R, R]^2$, Fubini and the exponential product law give

$$\iint_{Q_R} e^{-(x^2+y^2)} \, dA = \left(\int_{-R}^{R} e^{-x^2} \, dx \right) \left(\int_{-R}^{R} e^{-y^2} \, dy \right) = J_R^2.$$

The disc, square and larger disc satisfy $D_R \subset Q_R \subset D_{\sqrt{2}R}$. Indeed, a point of the first disc has each coordinate at most R in absolute value, and a point of the square has $x^2 + y^2 \le 2R^2$.

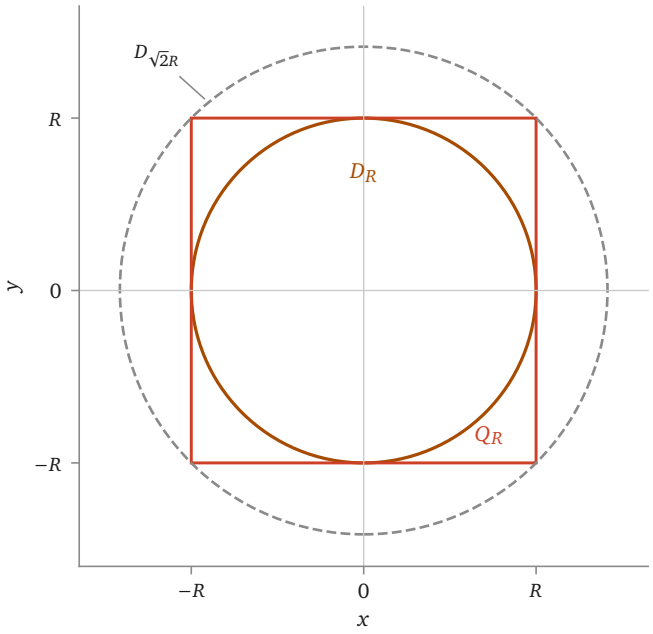


Figure 38.2: The disc D_R lies inside the square Q_R, which lies inside $D_{\sqrt{2}R}$. These are compact regions; integral order gives the Gaussian bounds.

The closure of $Q_R \setminus D_R$ consists of four corner graph pieces; that of $D_{\sqrt{2}R} \setminus Q_R$ consists of four circular segments of graph type. Their boundaries are piecewise C^1, so integral order applies to these inclusions. Figure 38.2 shows them.

Integral order for the nonnegative Gaussian and Eq. (38.2) yield

$$\pi(1 - e^{-R^2}) \le J_R^2 \le \pi(1 - e^{-2R^2}).$$

Lecture 27 already proves convergence of the two ordinary half-line Gaussian integrals. Their evenness makes $I = \int_{-\infty}^{\infty} e^{-x^2}\,dx = 2\int_0^{\infty} e^{-x^2}\,dx$ and $J_R \to I$ as $R \to \infty$. This two-sided notation denotes the sum of separately convergent halves, not a symmetric cancellation convention. Both bounds tend to π, so squeeze gives $I^2 = \pi$. The integral I is positive, since even its part on $[0, 1]$ is at least $e^{-1} > 0$. Therefore

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}, \qquad \int_0^{\infty} e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}, \qquad \Gamma(1/2) = \sqrt{\pi}. \tag{38.3}$$

The last equality uses Lecture 27's proved substitution identity. No improper Fubini theorem was used: Fubini and polar changes were applied only to the compact regions before the ordinary one-variable limit.

The normalisation of Lecture 27's error function now follows. Its definition is $\operatorname{erf} x = (2/\sqrt{\pi}) \int_0^x e^{-t^2}\,dt$. It is odd and its limits at $+\infty$ and $-\infty$ are 1 and -1. For $b > a$, its difference is the positive integral $(2/\sqrt{\pi}) \int_a^b e^{-t^2}\,dt$, so it is strictly increasing. Positivity is strict because, with $b_* = \max(|a|, |b|)$, the integrand is at least $e^{-b_*^2} > 0$ throughout that finite interval.

38.6 A proper parameter integral and its Taylor bound

The Gaussian calculation above used the weight e^{-r^2}. What changes if a parameter controls how rapidly this weight decreases from the origin? We first obtain a certified finite approximation to the resulting integral, then ask how its weighted mean squared radius changes

with the parameter. This also gives a concrete use of M's parameter-differentiation clause. The parameter clause of M applies before any radius tends to infinity. Fix $R > 0$ and define

$$F_R(\alpha) = \iint_{D_R} e^{-\alpha(x^2+y^2)}\,\mathrm{d}A \qquad (\alpha \in \mathbb{R}).$$

This is a proper integral for every real parameter, including negative ones. For each compact parameter interval the integrand is jointly C^1 on a neighbourhood of its product with the disc. All its higher parameter derivatives have the same property. Applying M successively therefore gives

$$F_R'(\alpha) = -\iint_{D_R} (x^2+y^2)e^{-\alpha(x^2+y^2)}\,\mathrm{d}A,$$

with higher derivatives obtained by further factors of $-(x^2+y^2)$. Polar integration also evaluates the function directly:

$$F_R(\alpha) = \begin{cases} \pi(1-e^{-\alpha R^2})/\alpha, & \alpha \neq 0, \\ \pi R^2, & \alpha = 0. \end{cases}$$

For nonzero α the radial primitive is $-e^{-\alpha r^2}/(2\alpha)$. At zero the integral is simply the disc area; division by the parameter is unnecessary. At $\alpha = 1$, this is precisely the disc integral used in the Gaussian squeeze above. For positive α, its limit as $R \to \infty$ will be π/α; the compact calculation comes first.

The differentiated-integral formulas give values at zero without a quotient limit. Since $2\pi \int_0^R r^{2j+1}\,\mathrm{d}r = \pi R^{2j+2}/(j+1)$, one obtains

$$F_R(0) = \pi R^2, \qquad F_R'(0) = -\frac{\pi R^4}{2}, \qquad F_R''(0) = \frac{\pi R^6}{3}.$$

For $\alpha \geq 0$, the third derivative on $[0,\alpha]$ lies between $-2\pi \int_0^R r^7\,\mathrm{d}r = -\pi R^8/4$ and zero. The scalar degree-2 Taylor formula consequently gives the signed bound

$$-\frac{\pi R^8 \alpha^3}{24} \leq F_R(\alpha) - \pi\left(R^2 - \frac{R^4\alpha}{2} + \frac{R^6\alpha^2}{6}\right) \leq 0.$$

At $\alpha = 0$ both errors are zero directly. For example, with $R = 1$ and $\alpha = 1/4$, the polynomial value is $85\pi/96$ and the error magnitude is at most $\pi/1536$. Thus

$$\frac{453\pi}{512} \leq F_1(1/4) \leq \frac{85\pi}{96}.$$

This certificate uses only proper parameter differentiation and a finite Taylor remainder. Although this example also has a closed form, the same method can certify an approximation when a parameter integral has no convenient closed form.

The parameter derivative also measures how the weight is distributed within this fixed disc. Write $q = x^2 + y^2$ and define its normalised radial moment by

$$\mu_R(\alpha) = \frac{\iint_{D_R} qe^{-\alpha q}\,\mathrm{d}A}{F_R(\alpha)} = -\frac{F_R'(\alpha)}{F_R(\alpha)}.$$

The denominator is positive, since the continuous weight is positive and the disc has positive area. Quotient differentiation and M give

$$\mu_R'(\alpha) = -\left(\frac{\iint_{D_R} q^2 e^{-\alpha q}\,\mathrm{d}A}{F_R(\alpha)} - \mu_R(\alpha)^2\right) = -\frac{\iint_{D_R} (q-\mu_R(\alpha))^2 e^{-\alpha q}\,\mathrm{d}A}{F_R(\alpha)} \leq 0.$$

To obtain the last identity, expand the square and use the definition of μ_R. The final inequality follows from integral order, without an additional probability theorem. Increasing the parameter therefore cannot increase the weighted mean of the squared distance from the origin. At zero parameter the preceding moments give $\mu_R(0) = R^2/2$ and $\mu_R'(0) = -R^4/12$. This interpretation concerns a fixed compact disc throughout.

For fixed $\alpha > 0$, the explicit formula gives $F_R(\alpha) \to \pi/\alpha$ as $R \to \infty$. For $\alpha = 0$ it grows as πR^2; for $\alpha < 0$ it equals $\pi(e^{|\alpha|R^2} - 1)/|\alpha|$ and also tends to infinity. Thus finiteness for every parameter on each compact disc does not imply finiteness for every parameter after taking the unbounded-radius limit.

38.7 Mass and centroid of a plate

For a thin plate on a compact region D, assume a continuous surface mass density $\rho(x, y) \geq 0$, measured as mass per unit area. Its small-area mass model gives total mass m and first moments

$$m = \iint_D \rho \, \mathrm{d}A, \qquad Q_x = \iint_D x\rho \, \mathrm{d}A, \qquad Q_y = \iint_D y\rho \, \mathrm{d}A.$$

Here Q_x means the first moment of the x coordinate, and Q_y that of the y coordinate; the subscript does not name an axis of rotation. If $m > 0$, the *centroid* is $(\bar{x}, \bar{y}) = (Q_x/m, Q_y/m)$, the continuous version of a mass-weighted mean. For uniform positive density it is also the centroid of the region's area.

Example 38.3 (A triangular plate with increasing density)**.** On $D = \{0 \leq y \leq x \leq 1\}$, let $\rho = \kappa(1 + x)$ with $\kappa > 0$. Find the mass and centroid. The compact Fubini calculation gives

$$m = \kappa \int_0^1 x(1 + x) \, \mathrm{d}x = \frac{5\kappa}{6}, \qquad Q_x = \kappa \int_0^1 x^2(1 + x) \, \mathrm{d}x = \frac{7\kappa}{12}.$$

For the other coordinate the inner primitive is $y^2/2$, giving

$$Q_y = \kappa \int_0^1 \frac{x^2}{2}(1 + x) \, \mathrm{d}x = \frac{7\kappa}{24}.$$

The mass is positive, so normalising yields $\bar{x} = 7/10$ and $\bar{y} = 7/20$. The centroid lies inside the triangle, since $0 < 7/20 < 7/10 < 1$.

Summary 38.

- Standing fact M supplies existence, Fubini, admissible coordinate changes and differentiation under proper compact integrals under its stated continuity and regularity hypotheses. It does not supply an improper multiple-integral rule.
- A change of order must describe the same region. An admissible coordinate change multiplies the integrand by $|\det J|$; polar area has factor r.
- Compact discs and squares give the squeeze leading to Eq. (38.3). The error function consequently has limits ± 1.
- Planar mass and first coordinate moments are double integrals. Their ratios give the centroid when the total mass is positive.

Sheet 38 treats curved regions, nonlinear coordinates, compact Gaussian cutoffs, plate centroids and the parameter clause of M. Lecture 39 extends the same compact rules to triple integrals and moments of solids.

References

V. A. Zorich, *Mathematical Analysis II*, 2nd English ed., Springer, 2016, Chapter 11 and §17.1 (standing fact M). J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§15.1–15.4 and §15.9.